
FLUOR DANIEL

CONVEYOR SUPPORT SYSTEMS**PURPOSE**

This practice establishes guidelines and recommended procedures for the design of conveyor support systems.

SCOPE

This practice describes the general arrangement of a belt conveyor support system and its various components, and gives design criteria for the design of such a system. It also provides material characteristics and weights per cubic foot for commonly conveyed bulk materials and other tables to facilitate design, including the following:

- Weights of conveyor belts.
- Idler weights.
- Recommended spacing for idlers.
- Maximum recommended belt speeds.
- Belt conveyor cross sectional area based on angle of surcharge.
- Friction factors.
- Maximum conveying incline angles.
- Sample design.

APPLICATION

This practice assumes that a belt conveyor system will be selected, and provides basic information to Structural Engineers regarding the belt conveyors and properties of commonly conveyed materials so that they may design the support system for the conveyors. The belt conveyors are adaptable for performing other functions besides conveying the materials. This includes weighing, blending, sampling, stock piling, sorting, and spraying.

GENERAL

Material handling is common to most industries served by Fluor Daniel. This includes metals and mining, stone, aggregate and construction industries, steel plants, coal fired power plants, incinerators, pulp and paper industry, food, chemical, and other process plants.

DESCRIPTION**Belt Conveyor**

The belt conveyor consists of 5 essential elements (refer to Attachment 08).

Belt

The belt consists of a closed moving belt loop that provides supporting surface for the material being conveyed. The belt travels over a head pulley on 1 end and a tail pulley at the other with idlers, snub pulley, and gravity takeup pulley in between.

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Idlers

Idlers form the supports for the troughed carrying strand of the belt and the flat return strand. Specified by Fluor Daniel, troughing and return idlers are generally of the roller bearing type. Troughing idlers are 4 to 7 inches in diameter and return idlers are a minimum of 4 inches in diameter. The return idlers are usually smaller in diameter than the carrying idlers.

Pulleys

Pulleys support and direct the belt and control the tension in the belt.

Drive

The drive imparts power through 1 or more pulleys to move the belt and its load. In most situations, the head pulley is used for driving.

Structure

The physical structure is designed to support and maintain the alignment of mechanical equipment of the conveyors (pulleys, idlers, drives, and belts).

Accessories

In addition to the essential elements, there are numerous accessories that add to the efficiency and increase the uses of a belt conveyor such as trippers, take-ups, chutes, feeders, belt cleaners, and holdbacks.

Tripper

Tripper may be described as a pair of bend pulleys located over the top run of a conveyor, for the purpose of discharging material before it reaches the head pulley. A chute can then be arranged at the upper bend pulley of tripper to discharge on either side of the conveyor. Trippers may either be stationary or traveling and the transverse motion can either be taken from the belt or tripper can be motor propelled.

Take-Up

Every belt conveyor should be equipped with a take-up. Its purpose is 3-fold, as follows:

- To allow for stretch or shrinkage of belt.
- To ensure that a minimum tension of the belt is sufficient to prevent undue sag between the idlers.
- To ensure that the tension of the belt back of drive pulley is sufficient to permit the pulley to transmit the load.

On short belts, screw take-ups are used at the tail pulley. The advantage is economy and compactness. The disadvantage is that the amount of tension put on the belt can not be checked.

On large belts, more than 100 to 150 feet long, gravity take-ups should be used, either the vertical or horizontal type.

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CONVEYOR SUPPORT SYSTEMS**Holdback**

Gravity take-up should be located on the return strand as close to the drive as possible, this may have to be altered with regard to the space available.

Holdbacks should be specified on inclined belt conveyors to prevent a fully loaded belt from running backwards in case of power failure.

Belt Cleaner

Belt cleaners should be specified at the discharge pulley.

Materials, particularly when wet, will have a tendency to stick to the belt and may drop off the return strand, thereby creating a cleanup problem unless rubber edge belt scrapers or rotating rubber brushes are specified at the head pulley.

Profiles

Attachment 02 illustrates commonly used profiles for the belt conveying system. The selection of the profile is dependent on the physical characteristics of the plant. The most important are the horizontal and vertical distances and the flowability of the materials to be transported. These factors determine the inclination of the conveyor. If the angle of inclination is steep, the material will roll back.

Support's

If a conveyor is located close to a floor level, its supports usually consist of a pair of channels carried on posts approximately 10-foot centers. When the conveyors are located above ground for overhead clearances or to direct the material to the top of storage bins or buildings, the most common conveyor supports are trusses carried on A-frames spaced 60 to 80 foot on centers.

Depth of open deck trusses in inches is approximately $1/12$ of the truss span in feet rounded up to 3-inch intervals. Chord members for typical truss are single angles for the open deck trusses. Weight of such a truss may be 100 pounds per foot. The platforms (walkway) and handrails are conventionally cantilevered from the truss.

Depth of gallery conveyor trusses is governed by the head room requirements. Typical depth is 8'-9". Weight of gallery truss is approximately 200 pounds per foot when conveyor belts are 4 feet or less in width. The weight of the truss may be as much as 300 pounds per foot when the belt is 5 feet or more in width.

Dust control and protection of personnel can be accomplished by enclosures. Enclosures could be limited only on the belt conveyors or they could be designed to totally enclose the conveyors, walkways, hand rails, and terminals. Housings prevent ice and wind from causing a belt to run off center and an empty belt from being blown off the idlers. They also decrease deterioration of the belt by protecting it from the sun. Typical galleries and housing are illustrated in Attachment 10.

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DESIGN CRITERIA

Structural supports for conveyors are to be designed for the following loads and load combinations:

DL (Dead Loads)

Self weight of structural members, hood covers and wind breakers.

Gravity take-up system weight: Take-up pulley is normally located near the upper section of the conveyor system but the counter weight is located in the ground. A platform is needed to service the pulley.

LL (Live Loads)

LL on walk ways:

- Design of the immediate support member 100 psf
- Analysis of trusses, bents, and columns 50 psf
- Concentrated load of 500 pounds on the beam.
- Weights of belt, troughing idlers, return idlers, and material conveyed on the belt (include estimate of spillage weight).
- Spacing of idlers, size, shape, and speed of belt are input to Structural Group. They are typically engineered by Mechanical Engineering or Contractor.

Belt Drag

Belt tension varies due to friction between pulleys, idlers, and belt. This force is generally small and neglected in the design of the supporting truss for the belt.

Tension

Belt tension force is generally taken by the head and tail pulley frames and their supporting structures. Tension force is an input to Structural group. It is computed by Mechanical Engineering group or Contractor. The design force is generally twice the maximum operating belt tension to account for the inertia forces of accelerations and startup or stoppage.

Wind

In accordance with job specifications (for example, Uniform or National Building Code, or ANSI A58.1).

Seismic

In accordance with job specifications (for example, Uniform Building Code, or National Building Code).

Note!!! Lateral forces due to wind or earthquake are taken by the horizontal trusses located at the top and bottom of the conveyor truss and transferred to truss end frames or bents.

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Temperature

Allow for thermal expansion and contraction of the truss members for temperature variations. Bottom chords will be subjected to frictional forces at the sliding and pin supports due to thermal movement of the truss. Assume coefficient of friction equals 0.15 at slide plates and 0.05 at pin supports.

Snow Or Ice

Snow or ice on exposed areas of hood and walkways based on geographic location of the site.

Load Combinations

Load combinations include the following:

- DL + LL
- DL + LL + Wind + Friction (temperature)
- DL + LL + Earthquake + Friction (temperature)
- DL + LL + Snow
- DL + LL + Friction (temperature)

Codes And Standards

Job criteria will specify the Codes and Standards for the project.

Allowable stresses may be increased by 1/3 when load combinations include wind or earthquake forces.

Deflections

The deflection should be checked with combined dead loads and live loads. The trusses of 80 feet or greater span should be cambered for approximately the dead load deflections. Deflections should be compared with AISC allowable.

OTHER CONSIDERATIONS

Initial layout of conveyor system is done during the general arrangement stage as a team effort between various groups (Civil, Structural, and Materials Handling Engineers, Client, and potential Contractor). Utility, economy, and appearance must be considered at this stage.

Width of the truss for open deck and conveyor support for the gallery truss is determined by providing a minimum clearance of 6 inches between the edge of the belt and the support. Walkway should have a minimum unobstructed clearance of 30 inches. The walkway should be located such that the vertical distance from the top of the belt to the top of the grating is 3'-6".

For gallery type, the depth of the truss should provide a minimum headroom clearance of 7'-6".

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REFERENCES

- AISC (American Institute for Steel Construction)
- ANSI (American National Standards Institute) A58.1
- CEMA (Conveyor Equipment Manufacturer's Association)
- CEMA Handbook
No. 102-1982 Conveyor Terms and Definitions
- NBC (National Building Code)
- UBC (Uniform Building Code)
- Conveyor and Idler Selection Procedure. Catalog of Hewitt-Robbins (Microfilm #7375-0315).
- Larsen, Donald H. The Design of a Coal Conveyor System to be Free Flowing, Freeze Proof, and Trouble Free Maintenance. Fourth Annual Coal Handling And Storage Symposium. Sponsored by Coal Mining and Processing and Industrial Presentations Inc. Jun. 1983.
- Materials Handling and Processing Equipment. Link-Belt Catalog 1000.
- Schulta, George A. Belt Conveyors. Chapter 23. Material Handling Handbook. 2nd Edition. Edited by Raymond A. Kulwiec.
- Wohlbier, R. H. Stacking Blending Reclaiming.
- Fluor Daniel Design Guides. Belt Conveyors And Feeders. Structural Engineering - Material Handling Mechanical Conveying. 1976.
- | | |
|--|-------------------------------------|
| Structural Engineering
Practice 670.215.1207: | Anchor Bolt Design Criteria |
| Structural Engineering
Practice 670.215.1208: | Base Plate Design Criteria |
| Structural Engineering
Practice 670.215.1209: | Bolted End Plate Moment Connections |
| Structural Engineering
Practice 670.215.1225: | Combined Foundations |
| Structural Engineering
Practice 000.215.5450: | Belt Conveyor Stringer And Details |
| Structural Engineering
Practice 000.215.5460: | Trussed Conveyors And Details |
| Structural Engineering
Practice 000.215.5470: | Gallery Conveyors And Details |
| Structural Engineering
Practice 000.215.5480: | Conveyor Bents And Details |

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MATERIAL CHARACTERISTICS AND WEIGHT PER CUBIC FOOT

Explanation of Codes

CATEGORY	MATERIAL CHARACTERISTICS		CODE
Size	Very Fine	100 mesh and under	A
	Fine	1/3 of an inch and under	B
	Granular	Under 1/2 of an inch	C
	Lumpy	Containing lumps over 1/2 of an inch	D
	Irregular	Stringy, interlocking, mats together	E
Flowability Angle of Repose	Very free flowing	5° angle of surcharge	1
	Free flowing	10° angle of surcharge	2
	Average flowing	20° of surcharge	3
	Sluggish	25° angle of surcharge	4
Abrasiveness	Nonabrasive		5
	Abrasive		6
	Very abrasive		7
	Very Sharp; cuts or gouges belt covers		8
Miscellaneous Characteristics (Sometimes more than one of these characteristics may apply.)	Very dusty		L
	Aerates and develops fluid characteristics		M
	Contains explosive dust		N
	Contaminable, affecting use or saleability		P
	Degradable, affecting use or saleability		Q
	Gives off harmful fumes or dust		R
	Highly corrosive		S
	Mildly corrosive		T
	Hygroscopic		U
	Interlocks or mats		V
	Oils or chemicals present; may affect rubber products		W
	Packs under pressure		X
	Very light and fluffy, may be wind swept		Y
	Elevated temperature		Z

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MATERIAL CHARACTERISTICS AND WEIGHT PER CUBIC FOOT

MATERIAL	AVERAGE WEIGHT (lbs. / ft ³)	ANGLE OF REPOSE (degrees)	RECOMMENDED MAXIMUM INCLINATION	CODE
Alumina	50-65	22	10-12	B27M
Aluminum Oxide	70-120			A17M
Aluminum Silicate	49			B35S
Aluminum Sulphate	54	32	17	C25
Ammonium Sulphate (granular)	45-58			*C26S
Arsenate of Lead (refer to Lead Arsenate)				
Asbestos Ore or Rock	81			D27R
Asphalt, crushed, 1/2 of an inch and under	45			C35
Bagasse	7-10			E45Y
Bakelite and similar plastics (powdered)	35-45			B25
Barite	180			D36
Bark, wood, refuse	10-20	45	27	E46Y
Bauxite, ground, dry	68	35	20	B26
Bauxite, mine run	80-90	31	17	D37
Bauxite, crushed 3" and under	75-85		20	D37
Bentonite, crude	35-40			D46X
Bentonite, 100 mesh and under	50-60		20	A26XY
Bonemeal	55-60			B36
Borax, 2" to 3" lumps	60-65			D36
Borax, 1-1/2" to 2" lumps	55-60			D36
Borax, 1/2" screenings	55-60			C36
Borax, fine	45-55		20-22	B26T
Boron	75			•
Brick, hard	125			D47Z
Brick, soft	100			D47
Carborundum, 3" and under	100			D27
Casein	36			B35
Cast Iron Chips	130-200			C46
Cement, Portland	94	39	20-23	A26M
Cement, Portland, aerated	60-75			A16M
Cement, rock (refer to Limestone)				
Cement, clinker	75-95	30-40	18-20	D37
Chips, paper mill	20-25			E45
Chips paper mill, softwood	12-30			E45
Chips, hogged, fuel	15-25			E45W
Chrome Ore (chromite)	125-140			D27
Cinders, blast furnace	57	35	18-20	*D37T

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MATERIAL	AVERAGE WEIGHT (lbs. / ft ³)		ANGLE OF REPOSE (degrees)	RECOMMENDED MAXIMUM INCLINATION	CODE
Cinders, coal	40		35	20	*D37T
Clay, calcined	80-100				B37
Clay, dry, fines	100-120		35	20-22	C37
Clay, dry, lumpy	60-75		35	18-20	D36
Clinker, cement (refer to Cement Clinker)					
Coal, anthracite, river or culm., 1/2" and under	60		35	18	B35TY
Coal, anthracite, sized	55-60		27	16	C26
Coal, bituminous, mined 50 mesh and under	50-54		45	24	B45T
Coal, bituminous, mined, and sized	45-55		35	16	*
Coal, bituminous, mined, run of mine	45-55		38	18	D35T
Coal, bituminous, stripping, not cleaned	50-60				D36T
Coal, lignite	40-45		38	22	D36T
Coke breeze, 1/4" and under	25-35		30-45	20-22	C37Y
concrete, cinder	90-100			12-30	D56
Copper Ore	120-150			20	*D27
Copper Ore, crushed	100-150			*	D27
Dolomite, lumpy	90-100			22	D26
Earth, as excavated, dry	70-80		35	20	B36
Earth, wet containing clay	100-110		45	23	B46
Emery	230				A27
Epsom Salts	40-50				B25
Feldspar, 1/2" screenings	70-85		38	18	B36
Feldspar, 1-1/2" to 3" lumps	90-110		34	17	D36
Feldspar, 200 mesh	100				
Fish Meal	35-40				B45W
Flue Dust, boiler house, dry	35-40				A17MTY
Fluorspar, 1/2" screenings	85-105				C46
Fluorspar, 1/2" to 3" lumps	110-120				D46
Foundry Sand, loose (refer to Sand)	80-90				B47
Foundry Refuse and old sand cores	70-100				D37Z
Garbage, household	50				*E45VW
Glass, Batch	80-100			20-22	D27Z
Grain	lbs / ft ³	lbs / Bu			
Barley	38	47			B15N
Buckwheat	100	124			B25N
Corn, Cracked	45-50	56-62			C25W
Oats	26	32			C25W

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MATERIAL CHARACTERISTICS AND WEIGHT PER CUBIC FOOT

MATERIAL	AVERAGE WEIGHT (lbs. / ft ³)		ANGLE OF REPOSE (degrees)	RECOMMENDED MAXIMUM INCLINATION	CODE
Rice, Rough	36	45			B25M
Rice, Hulled or Polished	45-58	56-60			B15
Rye	42-46	52-57			B15N
Soybeans, Whole	45-50	56-62			C26NW
Soybean Meal	40	50			B35T
Wheat	45-58	56-60			C25N
Granite, 1/2" screenings	80-90				C27
Granite, 1-1/2 to 3" lumps	85-90				D27
Granite, broken	95-100				D27
Graphite, flake	40				C25
Gravel, bank run	90-100		38	20	
Gravel, dry, sharp	90-100			15-17	D27
Gravel, pebbles	90-100		30	12	D36
Gypsum Dust, nonaerated	93				
Gypsum Dust, aerated	60-70		42	23	A36Y
Gypsum, 1/2" screening	70-80		40	21	C36
Gypsum, 1-1/2" to 3" lumps	70-80		30	15	D26
Limelite Ore	140-160				B27
Iron Borings, machine shop	125				D47WZ
Iron Ore	100-200		35	18-20	*D36
Iron Ore, crushed	135-150			20-22	*C26
Iron Oxide, pigment	25		40	25	A45
Kaolin Clay, 3" and under	63		35	19	D36
Lead Ores	200-270		30	15	*B36RT
Lead Oxides	60-150				B53
Lignite, air dries	45-55				*D25
Lime, ground, 1/2" and under	60-65		43	23	B45X
Lime, hydrated, 1/2 " and under	40		40	21	B35MX
Lime, hydrated, pulverized	32-40		42	22	A35MXY
Lime, pebble	53-56		30	17	D35
Limestone, agricultural, 1/2" and under	68			20	B26
Limestone, crushed	85-90		38	18	C26X
Limestone, dust	80-85			20	A46MY
Manganese Ore	125-140		39	20	*D37
Marble, crushed, 1/2" and under	80-95				D27
Meat Scraps	50-55				E35VW
Mica, ground	13-15		34	23	*B36

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MATERIAL CHARACTERISTICS AND WEIGHT PER CUBIC FOOT

MATERIAL	AVERAGE WEIGHT (lbs. / ft ³)	ANGLE OF REPOSE (degrees)	RECOMMENDED MAXIMUM INCLINATION	CODE
Mica, pulverized	13-15			A26MY
Mica flakes	17-22			B16MY
Mill Scale				E46T
Molybdenite, powdered	107	40	25	B25
Muriate of Potash	77			B27
Nickel, Cobalt Sulphate Ore	80-150			*D27T
Oil Cake	48-50			D45W
Oyster Shells ground under 1/2"	50-60			C36T
Oyster Shells, whole	80			D36TV
Paper Pulp Stock	40-60			*E15MV
Petroleum Coke (refer to Coke)				
Phosphate, acid, fertilizer	60	26	13	B25T
Phosphate, triple super, ground fertilizer	50-55	45	30	B45T
Phosphate Rock, broken, dry	75-85	25-30	12-15	D26
Phosphate Rock, pulverized	60	40	25	B36
Polystyrene Beads	40			B25
Potash Salts and sylvite	80			B25T
Pumice, 1/2" and under	40-45			B47
Pyrites, iron 2" to 3" lumps	135-145			D26T
Pyrites, pellets	120-130			C26T
Quartz, dust	70-80			A27Y
Quartz, 1/2" screenings	80-90			C27Z
Quartz, 1-1/2" to 3" lumps	85-95			D27Z
Rock, crushed	125-145			D26
Rock, soft, excavated with shovel	100-110		22	D36
Salt, common dry, coarse	40-55		18-22	C26TU
Salt, common dry, fine	70-80	25	11	D26TUW
Salt Cake, dry, coarse	85	36	21	B36TW
Salt Cake, dry, pulverized	60-85			B26NT
Saltpeter	80			--
Sand, bank, damp	110-130	45	20-22	B47
Sand, bank, dry	90-110	35	16-18	B37
Sand, foundry, prepared	80-90		24	B47
Sand, foundry, shakeout	90-100	39	22	D37
Sand, Silica, dry	90-100		10-15	B27
Sand, core	65	41	26	B45X
Sandstone, broken	85-90			D37

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MATERIAL CHARACTERISTICS AND WEIGHT PER CUBIC FOOT

MATERIAL	AVERAGE WEIGHT (lbs. / ft ³)	ANGLE OF REPOSE (degrees)	RECOMMENDED MAXIMUM INCLINATION	CODE
Sawdust	10-13	36	22	*B35
Sewage (sludge)	40-50			E25TW
Shale, broken	90-100			D26QZ
Shale, crushed	85-90	39	22	C36
Sinter	100-135			*D27
Slag, blast furnace, crushed	80-90	25	10	A27
Slag, furnace, granular, dry	60-65	25	13-16	C27
Slag, furnace, granular, wet	90-100	45	20-22	B47
Slate, dust	70-80	35	20	A36Y
Slate, crushed, 1/2" and under	80-90	28	15	C26
Slate, 1-1/2" to 3" lumps	85-95			D26
Steel Chips, crushed	100-150			D27WZ
Steel Trimmings	75-150	35	18	E37V
Sugar, granulated	50-55			B25PQT
Sugar, powdered	50-60			B35PTY
Sugar, raw, cane	55-65			B36TX
Sugar, wet, beet	55-65			B36TX
Sugar Cane, knifed	15-18			E45V
Sulphate, powdered	50-60		21	B25NW
Sulphate, crushed, 1/2" and under	50-60		20	C25NS
Sulphate, 3" and under	80-85		18	D25NS
Taconite, pellets	116-130		13-15	D17Q
Talc, powdered	50-60			A25MY
Talc, 1/2" screenings	80-90			C25
Talc, 1-1/2" to 3 inch lumps	85-95			D25
Talc, solid	165			--
Traprock, 1/2" screenings	90-100			C37
Traprock, 2" to 3" lumps	100-110			D37
Trisodium Phosphate	60			B36
Trisodium Phosphate, granular	60	26	11	B25
Trisodium Phosphate, pulverized	50	40	25	B35
Triple Super Phosphate	50-55			B36RS
Vermiculite, expanded	16			C35

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APPROXIMATE BELT WEIGHT

(pounds per lineal foot of belting)

Belt Width (inches)	Weight of Material Conveyed Pounds Per Cubic Foot		
	30 to 74 lbs.	75 to 129 lbs.	130 to 200 lbs.
14 -16 -18	3.1	4.0	4.1
24	4.2	5.7	6.2
30	5.3	7.2	8
36	9.2	9.6	11.5
42	10.7	11.5	13.8
48	13.6	14.2	16.6
54	15.3	16.9	19.3
60	17.7	19.4	21.4
60	19.3	21.5	23.6
72	21.4	23.6	25.5

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WEIGHT OF REVOLVING IDLER PARTS

Belt Width Inches	Weight per Idler, pounds											
	Troughed Belt Idlers											
	Series 1000	Series 1000	Series 3000 and Stringerflex						Series 4000			Uniflex
	4" steel rolls	4" steel rolls	5" steel rolls	5" steel rolls	6" steel rolls	6" gray iron rolls	5" rubber cushion rolls	6" rubber cushion rolls	6" steel rolls	7" steel rolls	7" rubber cushion rolls	3½" rubber tread rolls
14	14											
16	15	16	21									
18	16	17	22	24	28	46	21	25				6
20	17	18	23	26	30	51	22	26				
24	20	21	27	31	35	60	27	32				8
30	23	26	32	37	44	72	32	38				9
36		30	37	43	52	85	37	45	99	112	124	
42				51	59	100	43	52	113	127	140	
48				57	67	112	48	59	126	142	152	
54				63	75	125	54	66	140	157	174	
60				69	84	138	60	73	152	171	189	
72									178	198	220	

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WEIGHT OF REVOLVING IDLER PARTS

Weight of Revolving Idler Parts

Belt Width Inches	Weight per Idler, pounds												
	Flat belt and return Idlers												
	Series 1000	Series 2000	Series 3000 and Stringerflex								Series 4000		
	4" steel rolls	4" steel rolls	5" steel rolls	5" rubber tread rolls	4" steel rolls	5" steel rolls	6" rolls steel	6" gray iron rolls	5" rubber tread rolls	6" rubber tread rolls	6" steel rolls	7" steel rolls	6" rubber tread rolls
14	12												
16	13	14	17	7									
18	14	16	19	8	19	23	26	49	12	21			
20	15	17	21	9	21	25	29	52	13	22			
24	17	20	25	10	24	29	34	56	15	24			
30	21	24	30	12	30	35	42	68	18	28			
36		28	35	15	35	41	50	81	21	32	104	116	63
42					40	47	57	94	24	36	117	130	70
48					45	54	65	106	27	40	131	145	77
54					50	60	73	119	30	44	144	159	84
60					55	66	81	131	33	48	157	174	91
72											203	203	10

FLUOR DANIEL

NORMAL IDLER SPACING

Belt Width	Material Weight (lbs. / ft. ³)				
	30	50	100	130	200
18"	5'6"	5'6"	5'0"	4'9"	4'9"
24"	5'6"	5'6"	5'0"	4'6"	4'3"
30"	5'0"	5'6"	4'6"	4'3"	4'0"
36"	5'0"	5'0"	4'6"	4'3"	4'0"
42"	4'6"	4'6"	4'0"	3'9"	3'3"
48"	4'6"	4'0"	3'9"	3'6"	3'0"
54"	4'6"	4'0"	3'6"	3'3"	--
60"	4'0"	4'0"	3'0"	--	--

FLUOR DANIEL

MAXIMUM RECOMMENDED BELT SPEEDS

Material Being Conveyed	Belt Speeds (FPM)	Belt Width (inch)
Grain of other free-flowing, nonabrasive material	400 to 500 600 to 700 800	up to 24 24 to 36 Over 36
Coal, damp clay, soft ores, overburden and earth, fine crushed stone	300 to 400 500 to 600 700	up to 24 24 to 36 Over 36
Heavy, hard, sharp edged ore, coarse crushed stone	250 to 350 400 to 500 600	up to 24 24 to 36 Over 36
Shakeout foundry sand with small core lumps but no castings, gates, or sprues	150 to 200	Any width
Shakeout foundry sand mixed with small castings, gates, and sprues, not hot enough to injure belt	150 to 200	Any width
Prepared or damp foundry sand where no plows are used	250 to 350	Any width
Prepared foundry sand and similar damp (of dry abrasive) materials discharged from belt by rubber-edged plows	150 to 175	Any width
Nonabrasive materials discharged from belt by means of plows	200 maximum, except for wood pulp, where 300 to 400 is preferable	Any width
Feeder belts, flat or troughed, for feeding fine nonabrasive or mildly abrasive materials from hoppers and bins	50 to 100	Any width
Belts with trippers driven by the conveyor belt itself.	400 maximum	Any width

Note!!! The most economical selection usually dictated the use of maximum belt speeds.

FLUOR DANIEL

BELT CONVEYOR CROSS SECTIONAL AREA

Belt Width	5° SURCHARGE			10 ° SURCHARGE			20 ° SURCHARGE		
	20 °	35 °	45 °	20 °	35 °	45 °	20 °	35 °	45 °
14"	0.06398	0.09427	0.10786	0.07476	0.10306	0.11593	0.09681	0.12266	0.13241
18"	0.11531	0.16973	0.19385	0.13347	0.18622	0.20804	0.17382	0.21993	0.23705
20"	0.14659	0.21568	0.24618	0.17083	0.23652	0.26408	0.22038	0.279	0.30066
24"	0.22036	0.32404	0.36951	0.25655	0.35508	0.39612	0.33051	0.41582	0.45019
30"	0.35906	0.52769	0.60118	0.41766	0.57785	0.64408	0.53741	0.68034	0.73174
36"	0.53142	0.78068	0.88887	0.61780	0.84542	0.95193	0.79430	1.005	1.0808
42"	0.73743	1.083	1.233	0.85696	1.185	1.3196	1.101	1.394	1.498
48"	0.97710	1.435	1.632	1.135	1.569	1.7473	1.458	1.846	1.982
54"	1.25042	1.836	2.088	1.45236	2.008	2.235	1.865	2.360	2.535
60"	1.557	2.286	2.600	1.808	2.501	2.7822	2.322	2.940	3.155
72"	2.272	3.335	3.791	2.638	3.647	4.0566	3.39	4.284	4.599
84"	3.122	4.58	5.207	3.624	5.008	5.571	4.65	5.882	6.314
96"	4.106	8.024	6.847	4.766	6.586	7.324	8.113	7.732	8.229

Note!!! For good conveyer practice, size belt to handle 80 percent of peak tonnage.

FLUOR DANIEL

BELT CONVEYOR CROSS SECTIONAL AREA

Belt Width	25 ° SURCHARGE			30 ° SURCHARGE		
	20 °	35 °	45 °	20 °	35 °	45 °
14"	0.10821	0.13252	0.14093	0.11997	0.14268	0.14973
18"	0.19387	0.23736	0.25205	0.21476	0.25533	0.28752
20"	0.24598	0.30112	0.31958	0.2724	0.32383	0.3391
24"	0.36878	0.45132	0.47882	0.40821	0.48516	0.50762
30"	0.59933	0.73335	0.77707	0.66321	0.78802	0.82383
36"	0.88588	1.08342	1.1474	0.97974	1.1639	1.21615
42"	1.2275	1.50154	1.5896	1.3578	1.6128	1.68457
48"	1.6251	1.98772	2.10377	1.7974	2.1347	2.2910
54"	2.040	2.498	2.656	2.2985	2.7297	2.8497
60"	2.5874	3.16424	3.3476	2.8612	3.3977	3.5465
72"	3.7724	4.61296	4.8791	4.1712	4.9527	5.1683
84"	6.18	6.333	6.698	5.7273	6.7999	7.0945
96"	8.81	8.327	8.805	7.5295	8.9392	9.3251

Note!!! For good conveyor practice, size belt to handle 80 percent of peak tonnage.

FLUOR DANIEL

MAXIMUM CONVEYING INCLINE ANGLES

Material Carried	Maximum Angle of Incline ^a (degrees)	Material Carried	Maximum Angle of Incline ^a (degrees)
Alumina, dry, free flowing	18	Ore (refer to stone)	15-20
Beans, whole	8	Packages	15-25
Coal, anthracite	16	Pellets, depending on size, bed of material and concentricity (taconite and fertilizer)	5-15
Coal, bituminous, sized, lumps over 4 inches	15	Rock (see stone)	15-20
Coal, bituminous, sized, lumps 4 inches and under	16	Sand, very free flowing ^a	15
Coal bituminous, unsized	18	Sand, sluggish (moist) ^c	20
Coal, bituminous, fines, free flowing ^b	20	Sand, tempered foundry	24
Coal, bituminous, fines, sluggish ^c	22	Stone, sized, lumps over 4 inches	15
Coke, sized	17	Stone, sized, lumps 4 inches and under 3/8 of an inch	16
Coke, unsized	18	Stone, unsized, lumps over 4 inches	16
Coke, fines and breeze	20	Stone, unsized, lumps 4 inches and under	18
Earth, free flowing ^b	20	Stone, fines 3/8 of an inch and under	20
Earth, sluggish ^c	22	Wood chips	27
Gravel, sized, washed	12		
Gravel, sized, unwashed	15		
Gravel, unsized	18		
Grain	15		

^a For ascending conveyors when uniformly loaded and with constant feed.

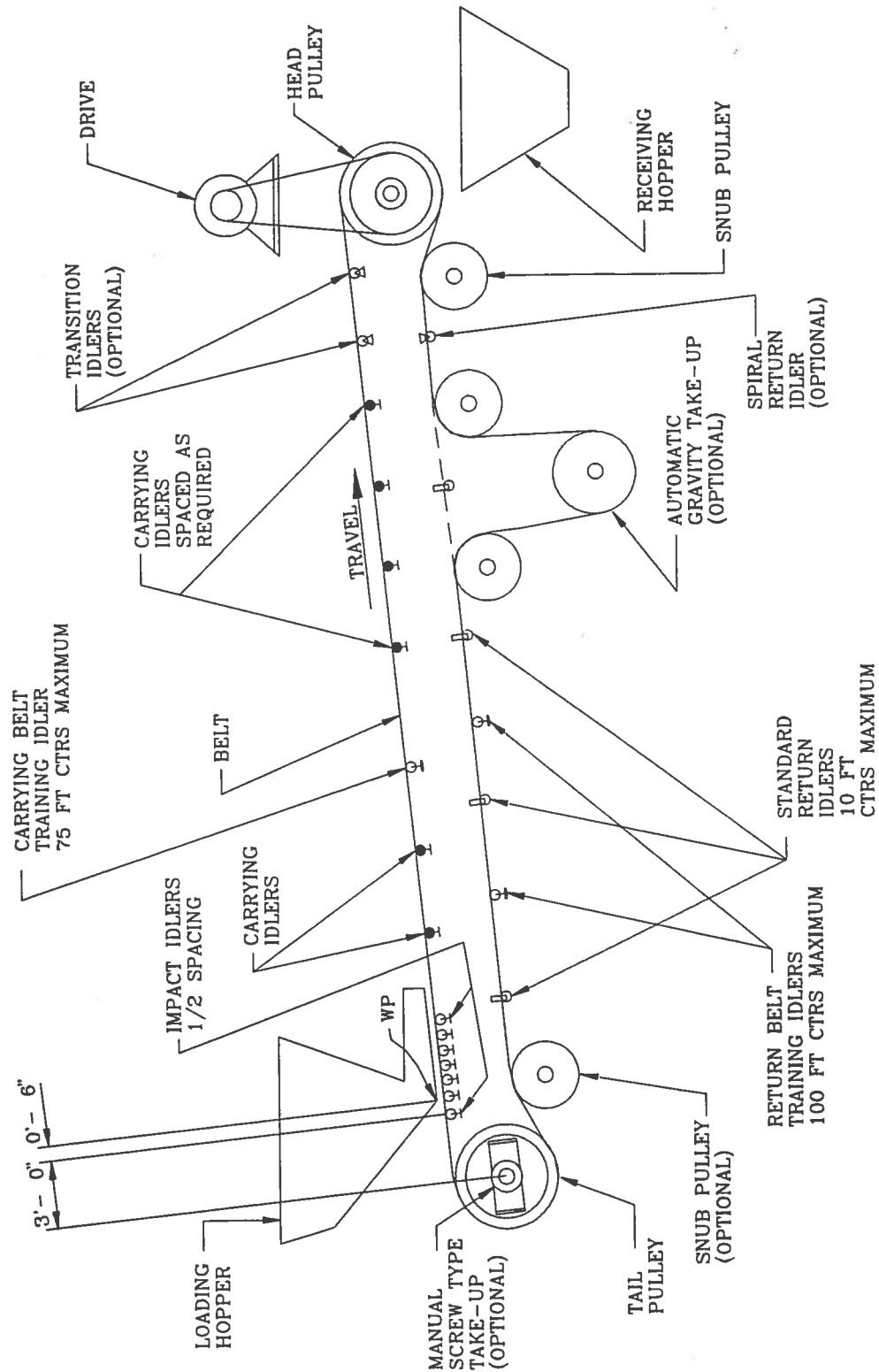
^b Angle of repose 30 to 45 degrees.

^c Angle of repose over 45 degrees.

^d Very wet or very dry, with angle of repose less than 30 degrees.

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BASIC BELT CONVEYOR COMPONENTS



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TYPICAL BELT CONVEYOR PROFILES



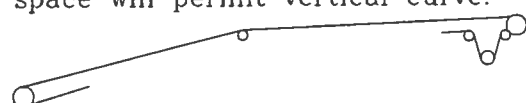
Horizontal Belt



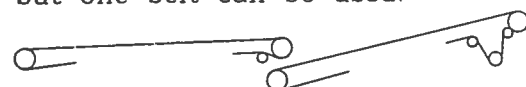
Horizontal and ascending path, when space will permit vertical curve and belt strength will permit one belt.



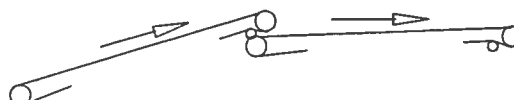
Ascending and horizontal path: when belt tensions will permit one belt and space will permit vertical curve.



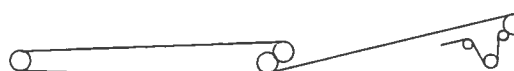
Ascending and horizontal path: or horizontal and descending path, when space will not permit vertical curve but one belt can be used.



Possible horizontal and ascending path, when space will not permit a vertical curve or when the conveyor belt strength requires 2 belts.



Ascending and horizontal path, when advisable to use 2 conveyor belts.



Possible horizontal and ascending path, when space will not permit vertical curve belt strength will permit 1 belt.



Compound path with declines, horizontal portions, vertical curves, and incline.



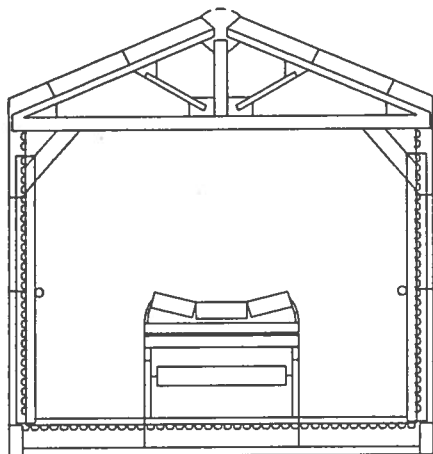
Loading can be accomplished, as shown on minor inclines or declines.



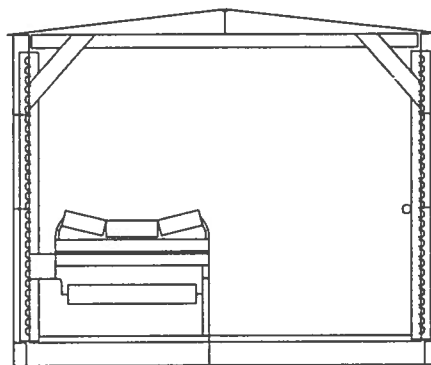
Traveling loading chute to receive materials as a number of points along conveyor.

FLUOR DANIEL

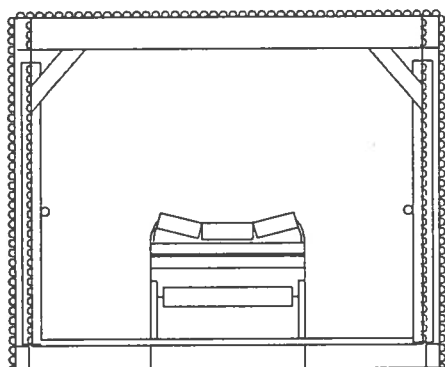
TYPICAL GALLERIES AND HOUSINGS



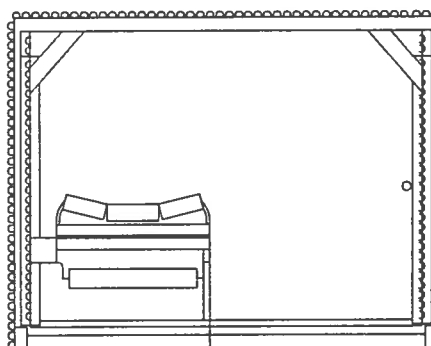
(A) Horizontal Closed Gallery with Walkway Both Sides



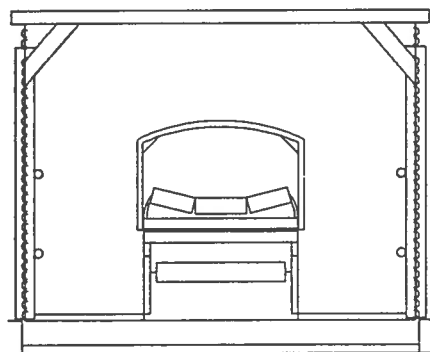
(B) Horizontal Closed Gallery with Walkway 1 Side



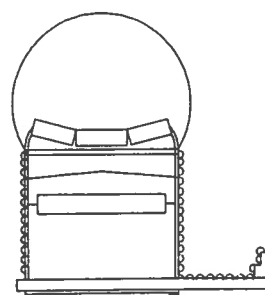
(C) Inclined Closed Gallery with Walkway Both Sides



(D) Inclined Closed Gallery with Walkway 1 Side



(E)



(G)

FLUOR DANIEL

SAMPLE DESIGN 1: OPEN DECK CONVEYOR TRUSS

OBJECTIVE : Design vertical and horizontal trusses for an open deck conveyor with the following information:

Material: Iron ore, density 110 pcf

Conveyor slope: 10°

Horizontal span = 43'

Walkway: on one side

Wind loading: 20 psf (UBC)

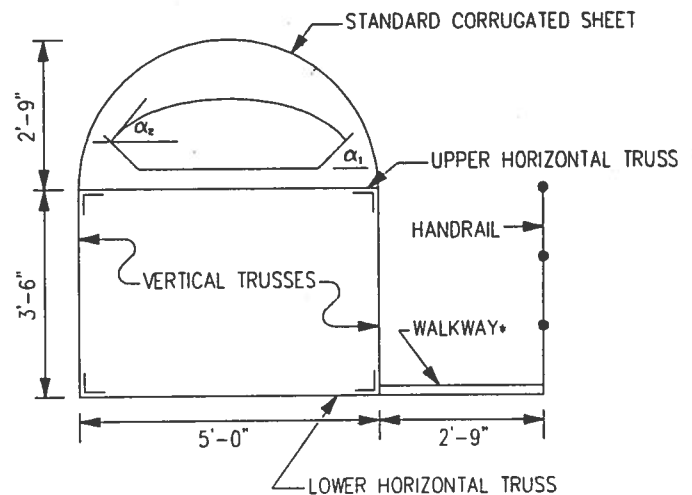
Belt width = 48", trough angle $35^\circ (\alpha_1)$

Note: From attachment 1,

- (a) Recommended maximum angle of inclination is $18-20^\circ > 10^\circ$ (okay)
- (b) Angle of repose for iron ore is 35° .
The material is classified as D36 meaning:

- (1) It is lumpy - contains lumps over $\frac{1}{2}$ "
- (2) It is average flowing, angle of surcharge will approximate $20^\circ (\alpha_2)$
- (3) It is abrasive material.

*For belt width ≥ 54 ", walkway is required on both sides of the belt.



VERTICAL TRUSS DESIGN

Select depth of truss = $\frac{1}{12} \times 43' \cong 43'' \rightarrow$ say 3' - 6"

Component Dead Loads

Weight of Ore = $1.846 (110) = 203 \text{ lb / ft}$

(From Attachment 6, with $\alpha_1 = 35^\circ$, $\alpha_2 = 20^\circ$, and belt width of 48", belt cross sectional area = 1.846 ft^2)

Weight of return idlers (assume 6" steel, series 3000)

Spacing of the return idlers is large as it carries weight of the belt only.

Assume spacing = 10'

Weight / ft = $\frac{67}{10} = 6.7 \text{ lb/ft} \cong 7 \text{ lb / ft}$

Weight of Belt: 14.2 lb/ft (from Attachment 2) $\cong 15 \text{ lb/ft}$

FLUOR DANIEL

SAMPLE DESIGN 1: OPEN DECK CONVEYOR TRUSS

Estimated weight of trusses = 200 lb/ft, 100 lb/ ft of vertical truss

Weight of walkway, handrail,etc = 35 lb/ft

Weight of 10 gage deck with 3" x 3" x $\frac{1}{4}$ " angle stiffeners @ 5' spacing = 35 lb/ft

Weight of corrugated galvanized hood (16 gage) = 35 lb/ft

Loads Summary:

Upper Panel Points of Truss: P_1

DL:

Idlers	36 / 2	= 18
Return Idlers	7 / 2	= 4
belts	15 / 2	= 8
truss	100 / 2	= 50
hood	33 / 2	= 17

LL:

Weight of Ore	205 / 2	= 102
	199 lb/ft	$\cong$ 200 lb/ft
P_1	$= 200 \times 5.375 = 1075 \text{ lb}$	

Lower Panel Points: P_2

DL:

Truss	100 / 2	= 50
Deck Plate	35 / 2	= 18

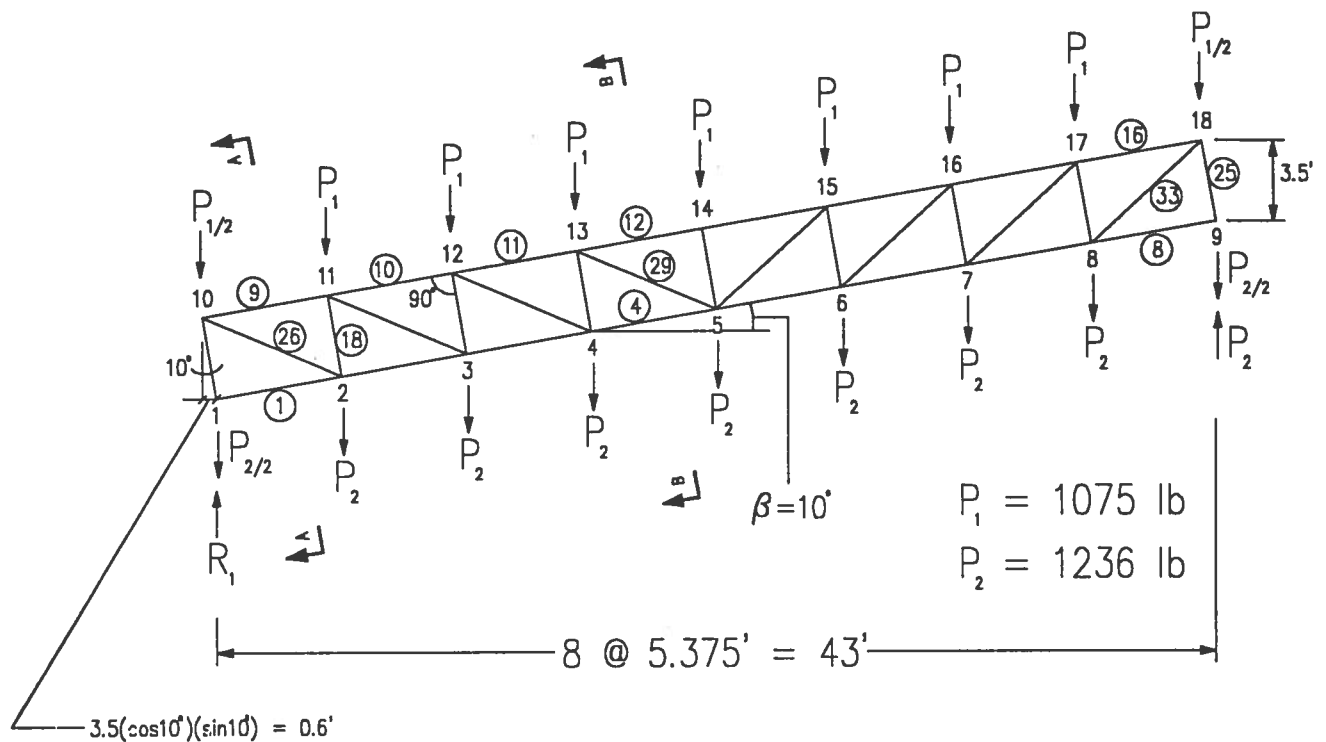
LL*:

Walkway, handrail, etc	$(50)(2.5)(6.4) / 5$	= 160
	228 lb/ft	$\cong$ 230 lb/ft
P_2	$= 230 \times 5.375 = 1236 \text{ lb}$	

*Conservative design : Entire walkway is not loaded at any one time

FLUOR DANIEL

SAMPLE DESIGN 1: OPEN DECK CONVEYOR TRUSS



$$\text{Length of members } \textcircled{1} - \textcircled{16} = 5.375 / \cos 10 = 5.46'$$

$$\text{Length of members } \textcircled{17} - \textcircled{25} = 3.5(\cos 10) = 3.45'$$

$$\text{Length of members } \textcircled{26} - \textcircled{33} = \sqrt{(5.46)^2 + (3.45)^2} = 6.46'$$

$$R_1 = 4(1075) + 4(1236) + 8(1075) \times 0.6 / 43 = 9365 \text{ lb}$$

$$R_2 = 4(1075) + 4(1236) - 8(1075) \times 0.6 / 43 = 9125 \text{ lb}$$

Use method of sections and method of joints equilibrium or use computer program such as RISA-2D to compute member forces.

Results for the design example are obtained by RISA-2D

Method of sections and joints equilibrium was used to check critical member forces.

FLUOR DANIEL

SAMPLE DESIGN 1: OPEN DECK CONVEYOR TRUSS

Consider equilibrium of truss left of section B-B:

For force F_{12} in member 12, take moment about Joint 5 :

$$\left[\left(9365 - \frac{1236}{2} \right) (21.5) - 1236(5.375 + 10.75 + 16.125) - 1075 \left(\frac{21.5 + 0.6}{2} + 5.375 + 0.6 + 10.75 + 0.6 + 16.125 + 0.6 \right) \right] + F_{12}(3.45) = 0$$

$$F_{12} = 28,903 \text{ lb} \cong 28.8 \text{ k (okay)}$$

For force F_4 in member 4, take moment about 13 :

$$\left(9,365 - \frac{1,236}{2} \right) [(3)(5.375) - 0.6] - 1236(10.75 - 0.6 + 5.375 - 0.6) + 1,236(0.6) - 1,075(5.375 + 10.75 + \frac{16.125}{2}) - F_4(3.45) = 0$$

$$F_4 = 26,692 \text{ lb} \cong 26.6 \text{ k (okay)}$$

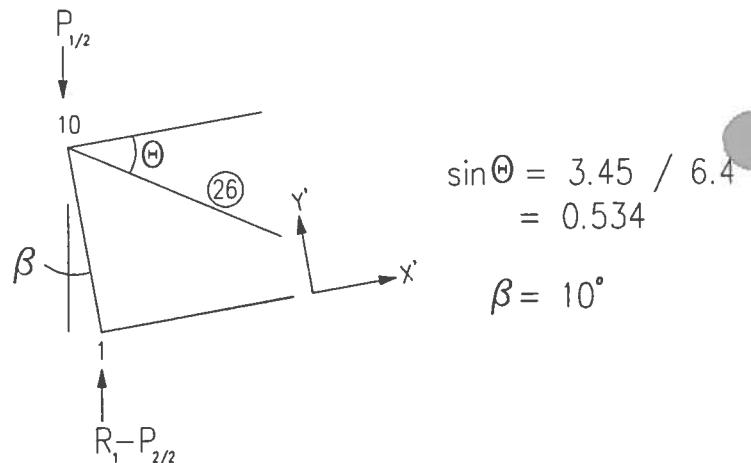
For force F_{26} in member 26 :

Consider equilibrium of forces on section left of A-A.

$$\Sigma F_{y'} = 0$$

$$F_{26} \sin \theta + \frac{P_1}{2} \cos \beta - (R_1 - \frac{P_2}{2}) \cos \beta = 0$$

$$F_{26} = 15,140 \text{ lb} \cong 15.1 \text{ k (okay)}$$

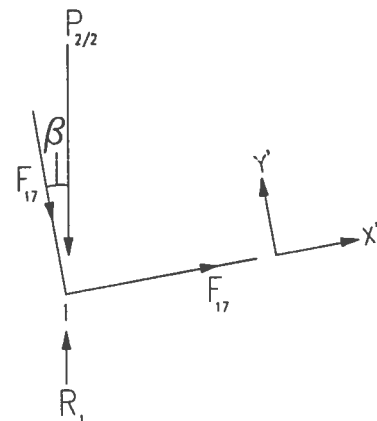


For computing force in member 17, consider equilibrium of joint 1 :

$$\Sigma F_{y''} = 0$$

$$F_{17} + \frac{P_2}{2} \cos \beta - R_1 \cos \beta = 0$$

$$F_{17} = 8,614 \text{ lb} \approx 8.6 \text{ k}$$



FLUOR DANIEL

SAMPLE DESIGN 1: OPEN DECK CONVEYOR TRUSS

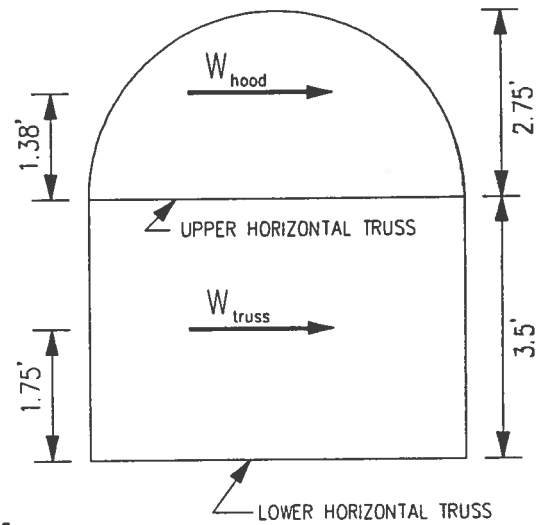
Wind Loads

$$W_{\text{hood}} = (20 \text{ psf})(2.75)(1.2) = 66 \text{ lb/ft}$$

Shape factors: 0.8 windward side
0.4 leeward side
1.2 total

$$W_{\text{truss}} = (20)(2.2)(1.5) = 66 \text{ lb/ft}$$

Estimate of vertical trusses' projected cross sectional area
Shape factor (2.2, UBC, Table 23-H)



$$\text{Wind on top of horizontal truss chord members} = [66(\frac{2.75}{2} + 3.5) + 66(\frac{3.5}{2})] / 3.5 = 125 \text{ lb/ft}$$

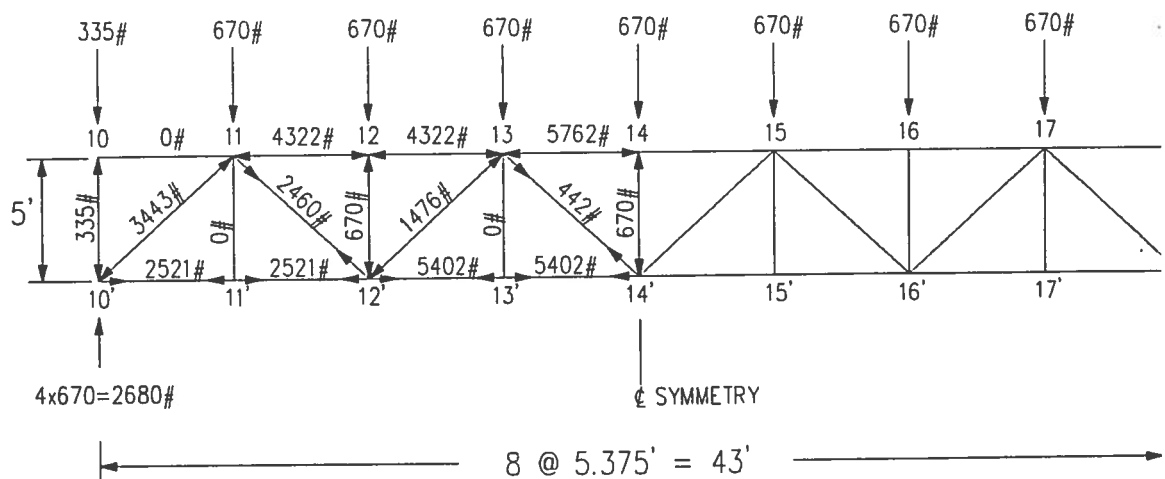
$$\text{Wind Load / Panel point} = (125)(5.375') = 670 \text{ lb/ft}$$

The horizontal wind loads will be carried to supports by the vertical members (bents, A-frame, truss) at the ends.

Top Horizontal Truss

Wind Loads

Compute member forces for the truss members (due to wind) by using computer program such as RISA 2D or by method of joints and/or sections.



- NOTE: 1. Wind loads are reversible
2. For the bottom truss, assume wind loads are 50% of the top truss members
3. Horizontal reactions will be carried by the vertical bents (A-frames) at the ends of truss.

FLUOR DANIEL

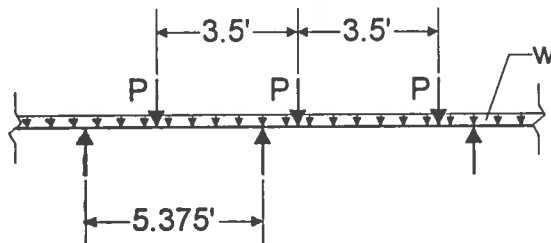
SAMPLE DESIGN 1: OPEN DECK CONVEYOR TRUSS

Design

Chord Members (common for vertical and horizontal forces)

Loa ding			
MEMBER	DL + LL	DL + LL + WIND	3/4(DL + LL + WIND)
<u>Top Chord</u> Member 12	28.84K (C)	28.84 + 3.76 = 32.6K (C)	25.95K (C)
<u>Bottom Chord</u> Member 5	27.25K (T)	27.25 + 5.40 = 32.65K (T)	24.49K (T)
<u>Vertical</u> Member 17	8.6K (C)	8.6K (C)	6.45K (C)
<u>Diagonal of Vertical Truss</u> Member 26	15.1K (T)	15.1K (T)	11.33K (T)
<u>Horizontal Truss</u> Diagonal	0	3.44K (C)	2.58K (C)

For the design of the top chord member include bending effects also due to self weight, location of troughing idler.



$$\begin{aligned}
 P &= (\text{weight of idler}) + (\text{load on idler}) \\
 &= 3.5' \times \frac{(15 + 203)}{2} + \frac{26}{2} \\
 &= 445 \text{ lbs}
 \end{aligned}$$

$$\begin{aligned}
 w &= (\text{weight of chord} + \text{weight of hood}) / \text{ft} \\
 &= 12 \text{ <assumed>} + \frac{33}{2} = 28.5 \text{ lbs / ft}
 \end{aligned}$$

$$M_{\text{center}} \cong \frac{PL}{8} + \frac{wL^2}{12} = 0.37 \text{ ft-k}$$

$$\text{Select } 6 \times 4 \times \frac{3}{8}'' \text{ (} Y_{\text{min}} = 0.877'', A = 3.61 \text{ in.}^2 \text{)}$$

$$L / r = 5.375 \times 12 / 0.877 = 73.5$$

$$f_t = 28.84 / 3.61 = 8 \text{ ksi}$$

$$f_b = 0.37 \times 12 / 3.32 = 1.34 \text{ ksi}$$

FLUOR DANIEL

SAMPLE DESIGN 1: OPEN DECK CONVEYOR TRUSS

Limiting Width / Thickness Ratio

$$\frac{b}{t} = \frac{95}{\sqrt{\frac{F_y}{k_c}}} = 15.83 \text{ (Table B5.1); where } F_y = 36, k_c = 1$$

$$\left(\frac{b}{t}\right)_{\text{selected}} = \frac{6}{\left(\frac{3}{8}\right)} = 16 \approx 15.83 \text{ (okay)}$$

From Appendix B of AISC (9th ed.)

$$Q_s = 1.34 - 0.00447 \frac{b}{t} \times \sqrt{F_y} = 0.91 \text{ (A-B5.1)}$$

Allowable Bending Stress

$$0.6 F_y \times Q_s = 19.66 \text{ ksi}$$

Allowable Axial Stress

$$F_a = \frac{Q[1 - \frac{(K/r)^2}{2C_c^2}]F_y}{\frac{5}{3} + \frac{3(K/r)}{8C_c} - \frac{(K/r)^3}{8C_c^2}} = 0.91 \times 16.06 = 14.6 \text{ ksi}$$

Use equations $H_{1.1}$ and $H_{1.2}$ to decide combined stress

$$C_m = 1.0$$

$$F_c = \frac{12\pi^2 E}{23(Kl_b/r_b)^2}; \text{ where } E = 29,000$$

$$l_b = 5.375 \times 12 = 64.5$$

$$r_b = 1.93$$

$$= 133.7$$

$$\frac{f_a}{F_a} + \frac{f_b}{(1 - \frac{f_a}{F_a})f_b} = \frac{8}{14.6} + \frac{1.34}{(1 - 8/133.7)19.66} = 0.62 \leq 1. \text{ (L } 6 \times 4 \times \frac{3}{8} \text{ are okay)}$$

Bottom Chord

Design Load = 27.25 k (Tension)

$$\text{Select L } 4 \times 3 \times \frac{1}{4} \text{ (A = 1.69 in.}^2, f_t = \frac{27.25}{1.69} = 16.1 \text{ ksi)}$$

$$F_t = 0.6F_y \quad \text{OR} \quad 0.5 F_u = 21.6 \text{ ksi} > 16.1 \text{ ksi per Section D1 AISC (L } 4 \times 3 \times \frac{1}{4} \text{ are okay)}$$

FLUOR DANIEL

SAMPLE DESIGN 1: OPEN DECK CONVEYOR TRUSS

Diagonals of Vertical Truss

$$P = 15.1 \text{ k (Tension)}$$

$$\text{Try } L \ 2\frac{1}{2} \times 2\frac{1}{2} \times \frac{1}{4} \ (A = 1.19 \text{ in.}^2)$$

$$f_t = 15.1 / 1.19 = 12.7 \text{ ksi} < 21.6 \text{ ksi (okay)}$$

Vertical Member of Truss

$$P = 8.6 \text{ k (Compression)}$$

$$\text{Use } L \ 2\frac{1}{2} \times 2\frac{1}{2} \times \frac{1}{4} \text{ (okay by inspection)}$$

Horizontal Truss (Diagonal)

$$P = 2.58 \text{ k (Compression)}$$

$$\text{Try } L \ 2\frac{1}{2} \times 2\frac{1}{2} \times \frac{1}{4} \ (\gamma_{\min} = 0.49, A = 1.19)$$

$$\frac{b}{t} = 10 > \frac{0.76}{\sqrt{F_y}}$$

$$Q_s = 1.0$$

$$L = \sqrt{5^2 + 5.375^2} \times 12 = 88"$$

$$\frac{KL}{r} = \frac{88}{0.49} = 180 < 200$$

$$F_a = 4.61 \text{ ksi}$$

$$f_a = \frac{2.58}{1.19} = 2.17 \text{ ksi} < 4.61 \text{ (okay)}$$

Design of Walkway Beam

Loads

$$\text{Weight of Hand rail + Toe Plate} = 35 \times 5.375 = 188 \text{ lb}$$

$$\text{Weight of Walkway} = 35 \times 5.375 = 188 \text{ lb}$$

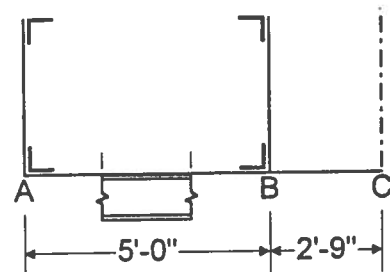
$$\text{Live Load on Walkway} = 100 \times 5.375 \times 2' - 9" = 1,478 \text{ lb}$$

$$M_b = 188 \times (2' - 9") / 2 + 1,478 \times (2' - 9") / 2 = 2,808 \text{ ft-lb} = 33.7 \text{ in.-k}$$

$$\text{Select } C \ 4 \times 5.4 \ (S = 1.93 \text{ in.}^3)$$

$$f_b = 33.7 / 1.93 = 17.46 \text{ ksi} < 0.6 \times 36 = 21.6 \text{ ksi (okay)}$$

RISA-2D (R) Version 3.03



FLUOR DANIEL

SAMPLE DESIGN 1: OPEN DECK CONVEYOR TRUSS

Fluor Daniel

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OPCON CONVEYOR VERTICAL TRUSS
PRACTICE 000 215 1258
=====

Units Option : US Standard

AISC Code Checks : No

Shear Deformation: No

P-Delta Effects : No

Redesign : No

Edge Forces : No

A.S.I.F. : 1.000

FLUOR DANIEL

SAMPLE DESIGN 1: OPEN DECK CONVEYOR TRUSS

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OPCON CONVEYOR VERTICAL TRUSS
PRACTICE 000 215 1258

Node No			Boundary Conditions			Temp. (F)
	X-Coord (ft)	Y-Coord (ft)	X-dof (in,K/in)	Y-dof (in,K/in)	Rotation (r,K-ft/r)	
1	0.00	0.00	R	R		0.00
2	5.38	0.93				0.00
3	10.75	1.87				0.00
4	16.13	2.80				0.00
5	21.50	3.73				0.00
6	26.88	4.67				0.00
7	32.25	5.60				0.00
8	37.63	6.54				0.00
9	43.00	7.47		R		0.00
10	-0.60	3.39				0.00
11	4.78	4.33				0.00
12	10.15	5.26				0.00
13	15.53	6.20				0.00
14	20.90	7.13				0.00
15	26.28	8.06				0.00
16	31.65	9.00				0.00
17	37.03	9.93				0.00
18	42.40	10.86				0.00

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SAMPLE DESIGN 1: OPEN DECK CONVEYOR TRUSS

RISA-2D (R) Version 3.03

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OPCON CONVEYOR VERTICAL TRUSS
PRACTICE 000 215 1258

Material Label	Elastic Modulus (Ksi)	Poisson's Ratio	Thermal Coefficient (F)	Weight Density (K/ft3)	Yield Stress (Fy) (Ksi)
STEEL	29000.00	0.30000	0.65000	0.490	36.000

RISA-2D (R) Version 3.03

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OPCON CONVEYOR VERTICAL TRUSS
PRACTICE 000 215 1258

Section Label	Database Shape	Matl. Set	Area (in^2)	Moment of Inertia (in^4)	As Coef	y/y
TOP CHORD		STEEL	3.61	1.000	1.20	
BOT CHORD		STEEL	1.69	1.000	1.20	
VERTICAL		STEEL	1.19	1.000	1.20	
DIAGONAL		STEEL	1.19	1.000	1.20	

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SAMPLE DESIGN 1: OPEN DECK CONVEYOR TRUSS

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OPCON CONVEYOR VERTICAL TRUSS
PRACTICE 000 215 1258

I		J		I Releases J						End Offsets		Length			
No	Node	Node	Section	x	y	z	x	y	z	Sec	Sway		I	J	
-----													(in)-----	(in)-----	(ft)
1	1	-	2	BOT CHORD											5.46
2	2	-	3	BOT CHORD											5.45
3	3	-	4	BOT CHORD											5.46
4	4	-	5	BOT CHORD											5.45
5	5	-	6	BOT CHORD											5.46
6	6	-	7	BOT CHORD											5.45
7	7	-	8	BOT CHORD											5.46
8	8	-	9	BOT CHORD											5.45
9	10	-	11	TOP CHORD											5.46
10	11	-	12	TOP CHORD											5.45
11	12	-	13	TOP CHORD											5.46
12	13	-	14	TOP CHORD											5.45
13	14	-	15	TOP CHORD											5.46
14	15	-	16	TOP CHORD											5.45
15	16	-	17	TOP CHORD											5.46
16	17	-	18	TOP CHORD											5.45
17	1	-	10	VERTICAL			1			1					3.44
18	2	-	11	VERTICAL			1			1					3.45
19	3	-	12	VERTICAL			1			1					3.44
20	4	-	13	VERTICAL			1			1					3.45
21	5	-	14	VERTICAL			1			1					3.45
22	6	-	15	VERTICAL			1			1					3.44
23	7	-	16	VERTICAL			1			1					3.45
24	8	-	17	VERTICAL			1			1					3.44
25	9	-	18	VERTICAL			1			1					3.44
26	10	-	2	DIAGONAL			1			1					6.47
27	11	-	3	DIAGONAL			1			1					6.46
28	12	-	4	DIAGONAL			1			1					6.47
29	13	-	5	DIAGONAL			1			1					6.46
30	5	-	15	DIAGONAL			1			1					6.45
31	6	-	16	DIAGONAL			1			1					6.44
32	7	-	17	DIAGONAL			1			1					6.45
33	8	-	18	DIAGONAL			1			1					6.44

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SAMPLE DESIGN 1: OPEN DECK CONVEYOR TRUSS

RISA-2D (R) Version 3.03

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OPCON CONVEYOR VERTICAL TRUSS
PRACTICE 000 215 1258

BLC No.	Basic Load Case Description	Load Totals		
		Nodal	Point	Dist.
1	DEAD LOAD + LIVE LOAD	18		

RISA-2D (R) Version 3.03

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OPCON CONVEYOR VERTICAL TRUSS
PRACTICE 000 215 1258

Nodal Loads, BLC 1: DEAD LOAD + LIVE LOAD

Node Number	Global X	Global Y	Moment
	(K)	(K)	(K-ft)
1	0.000	-0.618	0.000
2	0.000	-1.236	0.000
3	0.000	-1.236	0.000
4	0.000	-1.236	0.000
5	0.000	-1.236	0.000
6	0.000	-1.236	0.000
7	0.000	-1.236	0.000
8	0.000	-1.236	0.000
9	0.000	-0.618	0.000
10	0.000	-0.538	0.000
11	0.000	-1.075	0.000
12	0.000	-1.075	0.000
13	0.000	-1.075	0.000
14	0.000	-1.075	0.000
15	0.000	-1.075	0.000
16	0.000	-1.075	0.000
17	0.000	-1.075	0.000
18	0.000	-0.538	0.000

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SAMPLE DESIGN 1: OPEN DECK CONVEYOR TRUSS

RISA-2D (R) Version 3.03

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OPCON CONVEYOR VERTICAL TRUSS
PRACTICE 000 215 1258

Load Combination No.	Description	Self Wt Dir	BLC Fac	BLC Fac	BLC Fac	BLC Fac	BLC Fac	W E
								DYNA S V
1	DEAD + LIVE		1	1				

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OPCON CONVEYOR VERTICAL TRUSS
PRACTICE 000 215 1258

Load Combination is 1 : DEAD + LIVE
Nodal Displacements

Node	Global X (in)	Global Y (in)	Rotation (rad)
1	-0.00000	-0.00000	-0.00462
2	0.04776	-0.28823	-0.00414
3	0.10339	-0.51730	-0.00300
4	0.15620	-0.65878	-0.00144
5	0.19892	-0.69715	-0.00009
6	0.22891	-0.65675	0.00131
7	0.23424	-0.51364	0.00288
8	0.21292	-0.28537	0.00406
9	0.16550	-0.00000	0.00460
10	0.13509	0.01344	-0.00459
11	0.17721	-0.27375	-0.00410
12	0.20352	-0.50515	-0.00296
13	0.21194	-0.65172	-0.00141
14	0.20166	-0.69796	-0.00009
15	0.17831	-0.66825	0.00127
16	0.13806	-0.53591	0.00283
17	0.08638	-0.31580	0.00402
18	0.02969	-0.03422	0.00457

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SAMPLE DESIGN 1: OPEN DECK CONVEYOR TRUSS

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OPCON CONVEYOR VERTICAL TRUSS
PRACTICE 000 215 1258

Load Combination is 1 : DEAD + LIVE
Reactions

Node	Global X (K)	Global Y (K)	Moment (K-ft)
1	0.00000	9.36344	0.00000
9	0.00000	9.12556	0.00000
Totals	0.00000	18.48900	0.00000

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SAMPLE DESIGN 1: OPEN DECK CONVEYOR TRUSS

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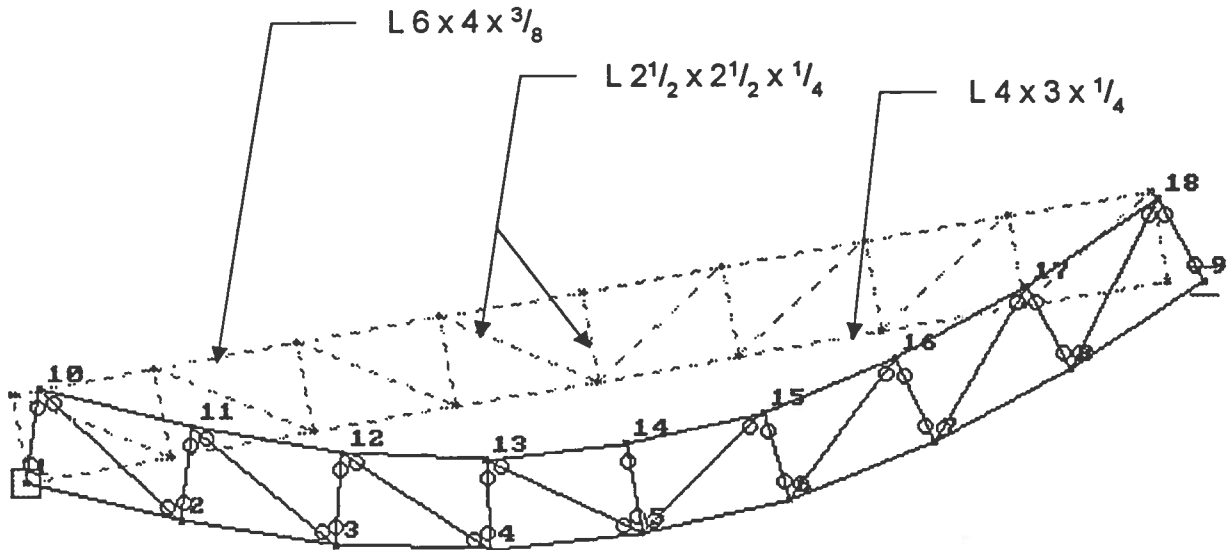
OPCON CONVEYOR VERTICAL TRUSS
PRACTICE 000 215 1258

Load Combination is 1 : DEAD + LIVE
Member End Forces

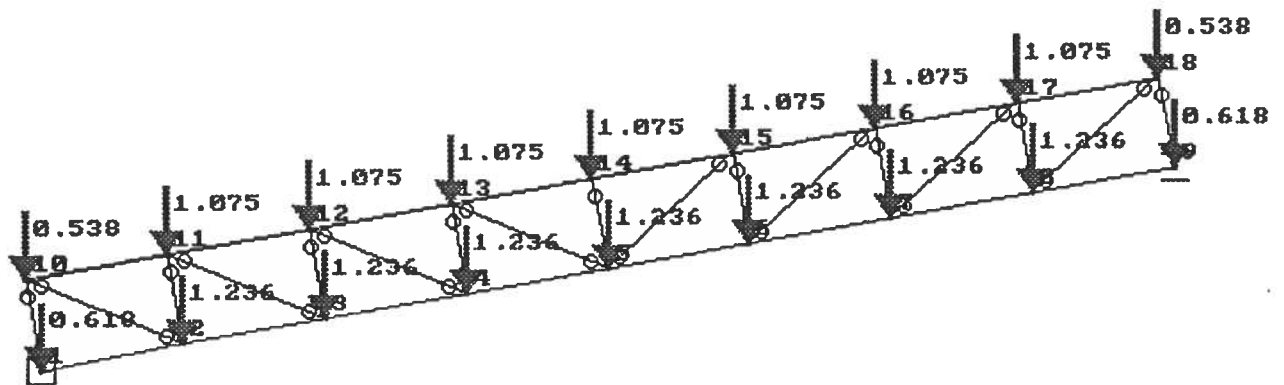
No	Nodes		I-End			J-End		
	I	J	Axial	Shear	Moment	Axial	Shear	Moment
			(K)	(K)	(K-ft)	(K)	(K)	(K-ft)
1	1-	2	1.52	0.01	0.00	-1.52	-0.01	0.04
2	2-	3	-11.46	0.00	-0.04	11.46	-0.00	0.05
3	3-	4	-20.90	0.00	-0.05	20.90	-0.00	0.07
4	4-	5	-26.63	-0.01	-0.07	26.63	0.01	0.03
5	5-	6	-27.29	0.01	-0.03	27.29	-0.01	0.07
6	6-	7	-22.23	-0.00	-0.07	22.23	0.00	0.05
7	7-	8	-13.68	-0.00	-0.05	13.68	0.00	0.04
8	8-	9	-1.48	-0.01	-0.04	1.48	0.01	-0.00
9	10-	11	12.68	0.01	-0.00	-12.68	-0.01	0.04
10	11-	12	21.71	0.00	-0.04	-21.71	-0.00	0.05
11	12-	13	27.04	0.00	-0.05	-27.04	-0.00	0.07
12	13-	14	28.86	-0.01	-0.07	-28.86	0.01	0.03
13	14-	15	28.67	0.01	-0.03	-28.67	-0.01	0.07
14	15-	16	26.92	-0.00	-0.07	-26.92	0.00	0.05
15	16-	17	21.47	-0.00	-0.05	-21.47	0.00	0.04
16	17-	18	12.50	-0.01	-0.04	-12.50	0.01	0.00
17	1-	10	8.61	0.00	0.00	-8.61	-0.00	0.00
18	2-	11	6.86	-0.00	0.00	-6.86	0.00	0.00
19	3-	12	4.58	-0.00	0.00	-4.58	0.00	0.00
20	4-	13	2.28	0.00	0.00	-2.28	-0.00	0.00
21	5-	14	1.06	-0.00	0.00	-1.06	-0.00	0.00
22	6-	15	2.09	-0.00	0.00	-2.09	0.00	0.00
23	7-	16	4.35	0.00	0.00	-4.35	-0.00	0.00
24	8-	17	6.61	0.00	0.00	-6.61	-0.00	0.00
25	9-	18	8.38	0.00	0.00	-8.38	-0.00	0.00
26	10-	2	-15.12	0.00	0.00	15.12	-0.00	0.00
27	11-	3	-10.93	-0.00	0.00	10.93	0.00	0.00
28	12-	4	-6.53	-0.00	0.00	6.53	0.00	0.00
29	13-	5	-2.37	-0.00	0.00	2.37	0.00	0.00
30	5-	15	-1.84	0.00	0.00	1.84	-0.00	0.00
31	6-	16	-6.23	0.00	0.00	6.23	-0.00	0.00
32	7-	17	-10.37	-0.00	0.00	10.37	0.00	0.00
33	8-	18	-14.65	-0.00	0.00	14.65	0.00	0.00

FLUOR DANIEL

SAMPLE DESIGN 1: OPEN DECK CONVEYOR TRUSS



Conveyor Vertical Truss
(Deflections Shown)



Conveyor Vertical Truss
Loading Diagram

FLUOR DANIEL

SAMPLE DESIGN 1: OPEN DECK CONVEYOR TRUSS

RISA-2D (R) Version 3.03

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OPCON CONVEYOR HORIZONTAL TRUSS
PRACTICE 000 215 1258
=====

Units Option : US Standard

AISC Code Checks : No

Shear Deformation: No

P-Delta Effects : No

Redesign : No

Edge Forces : No

A.S.I.F. : 1.000

FLUOR DANIEL

SAMPLE DESIGN 1: OPEN DECK CONVEYOR TRUSS

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OPCON CONVEYOR HORIZONTAL TRUSS
PRACTICE 000 215 1258

Node No	Boundary Conditions						Temp. (F)
	X-Coord (ft)	Y-Coord (ft)	X-dof (in,K/in)	Y-dof (in,K/in)	Rotation (r,K-ft/r)		
1	0.00	0.00	R	R			0.00
2	5.38	0.00					0.00
3	10.75	0.00					0.00
4	16.13	0.00					0.00
5	21.50	0.00					0.00
6	26.88	0.00					0.00
7	32.25	0.00					0.00
8	37.63	0.00					0.00
9	43.00	0.00			R		0.00
10	0.00	5.00					0.00
11	5.38	5.00					0.00
12	10.75	5.00					0.00
13	16.13	5.00					0.00
14	21.50	5.00					0.00
15	26.88	5.00					0.00
16	32.25	5.00					0.00
17	37.63	5.00					0.00
18	43.00	5.00					0.00

RISA-2D (R) Version 3.03

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OPCON CONVEYOR HORIZONTAL TRUSS
PRACTICE 000 215 1258

Material Label	Elastic Modulus (Ksi)	Poisson's Ratio	Thermal Coefficient (F)	Weight Density (K/ft3)	Yield Stress (Fy) (Ksi)
STEEL	29000.00	0.30000	0.65000	0.490	36.000

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SAMPLE DESIGN 1: OPEN DECK CONVEYOR TRUSS

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OPCON CONVEYOR HORIZONTAL TRUSS
PRACTICE 000 215 1258
=====

Section Label	Database Shape	Matl. Set	Area (in ²)	Moment of Inertia (in ⁴)	As Coef	y/y
CHORD		STEEL	3.61	1.000	1.20	
DIAGONAL		STEEL	1.19	1.000	1.20	
BRACES		STEEL	1.19	1.000	1.20	

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SAMPLE DESIGN 1: OPEN DECK CONVEYOR TRUSS

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OPCON CONVEYOR HORIZONTAL TRUSS

PRACTICE 000 215 1258

=====

I		J		Section	I Releases			J			End Offsets		Length	
No	Node	Node	x		y	z	x	y	z	Sec	Sway	I		J
												(in)	(in)	(ft)
1	1	-	2	CHORD										5.38
2	2	-	3	CHORD										5.37
3	3	-	4	CHORD										5.38
4	4	-	5	CHORD										5.37
5	5	-	6	CHORD										5.38
6	6	-	7	CHORD										5.37
7	7	-	8	CHORD										5.38
8	8	-	9	CHORD										5.37
9	10	-	11	CHORD										5.38
10	11	-	12	CHORD										5.37
11	12	-	13	CHORD										5.38
12	13	-	14	CHORD										5.37
13	14	-	15	CHORD										5.38
14	15	-	16	CHORD										5.37
15	16	-	17	CHORD										5.38
16	17	-	18	CHORD										5.37
17	1	-	10	BRACES			1				1			5.00
18	2	-	11	BRACES			1				1			5.00
19	3	-	12	BRACES			1				1			5.00
20	4	-	13	BRACES			1				1			5.00
21	5	-	14	BRACES			1				1			5.00
22	6	-	15	BRACES			1				1			5.00
23	7	-	16	BRACES			1				1			5.00
24	8	-	17	BRACES			1				1			5.00
25	9	-	18	BRACES			1				1			5.00
26	1	-	11	DIAGONAL			1				1			7.34
27	11	-	3	DIAGONAL			1				1			7.34
28	3	-	13	DIAGONAL			1				1			7.34
29	13	-	5	DIAGONAL			1				1			7.34
30	5	-	15	DIAGONAL			1				1			7.34
31	15	-	7	DIAGONAL			1				1			7.34
32	7	-	17	DIAGONAL			1				1			7.34
33	17	-	9	DIAGONAL			1				1			7.34

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SAMPLE DESIGN 1: OPEN DECK CONVEYOR TRUSS

RISA-2D (R) Version 3.03

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OPCON CONVEYOR HORIZONTAL TRUSS

PRACTICE 000 215 1258

BLC No.	Basic Load Case Description	Load Totals		
		Nodal	Point	Dist.
1	WIND LOAD	9		

RISA-2D (R) Version 3.03

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OPCON CONVEYOR HORIZONTAL TRUSS

PRACTICE 000 215 1258

Nodal Loads, BLC 1: WIND LOAD			
Node Number	Global X (K)	Global Y (K)	Moment (K-ft)
11	0.000	-0.670	0.000
12	0.000	-0.670	0.000
13	0.000	-0.670	0.000
14	0.000	-0.670	0.000
15	0.000	-0.670	0.000
16	0.000	-0.670	0.000
17	0.000	-0.670	0.000
10	0.000	-0.335	0.000
18	0.000	-0.335	0.000

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SAMPLE DESIGN 1: OPEN DECK CONVEYOR TRUSS

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OPCON CONVEYOR HORIZONTAL TRUSS

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Load Combination No.	Self Wt Dir	Wt Fac	BLC Fac	BLC Fac	BLC Fac	BLC Fac	BLC Fac	W DYN	E S
1 WIND			1	1					

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OPCON CONVEYOR HORIZONTAL TRUSS

PRACTICE 000 215 1258

Load Combination is 1 : WIND
Nodal Displacements

Node	Global X (in)	Global Y (in)	Rotation (rad)
1	-0.00000	-0.00000	-0.00053
2	0.00155	-0.03295	-0.00047
3	0.00311	-0.05882	-0.00033
4	0.00644	-0.07533	-0.00017
5	0.00976	-0.08097	0.00000
6	0.01309	-0.07531	0.00017
7	0.01642	-0.05882	0.00033
8	0.01797	-0.03288	0.00047
9	0.01952	-0.00000	0.00053
10	0.01863	-0.00058	-0.00051
11	0.01863	-0.03296	-0.00048
12	0.01597	-0.05998	-0.00033
13	0.01331	-0.07533	-0.00018
14	0.00976	-0.08213	0.00000
15	0.00621	-0.07531	0.00018
16	0.00355	-0.05998	0.00033
17	0.00089	-0.03289	0.00048
18	0.00089	-0.00058	0.00051

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SAMPLE DESIGN 1: OPEN DECK CONVEYOR TRUSS

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OPCON CONVEYOR HORIZONTAL TRUSS
PRACTICE 000 215 1258
=====

Load Combination is 1 : WIND
Reactions

Node	Global X	Global Y	Moment
	(K)	(K)	(K-ft)
1	0.00000	2.67969	0.00000
9	0.00000	2.68031	0.00000
Totals	0.00000	5.36000	0.00000

FLUOR DANIEL

SAMPLE DESIGN 1: OPEN DECK CONVEYOR TRUSS

RISA-2D (R) Version 3.03

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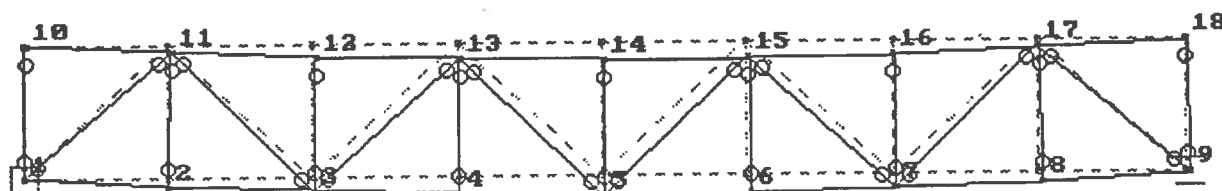
OPCON CONVEYOR HORIZONTAL TRUSS
PRACTICE 000 215 1258

Load Combination is 1 : WIND
Member End Forces

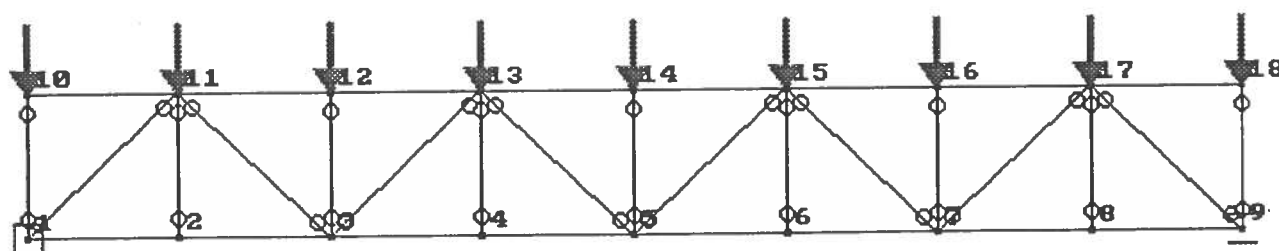
No	Nodes		I-End			J-End		
	I	J	Axial	Shear	Moment	Axial	Shear	Moment
			(K)	(K)	(K-ft)	(K)	(K)	(K-ft)
1	1-	2	-2.52	0.00	-0.00	2.52	-0.00	0.00
2	2-	3	-2.52	0.00	-0.00	2.52	-0.00	0.01
3	3-	4	-5.40	0.00	-0.01	5.40	-0.00	0.01
4	4-	5	-5.40	0.00	-0.01	5.40	-0.00	0.01
5	5-	6	-5.40	-0.00	-0.01	5.40	0.00	0.01
6	6-	7	-5.40	-0.00	-0.01	5.40	0.00	0.01
7	7-	8	-2.52	-0.00	-0.01	2.52	0.00	0.00
8	8-	9	-2.52	-0.00	-0.00	2.52	0.00	0.00
9	10-	11	0.00	0.00	0.00	0.00	-0.00	0.00
10	11-	12	4.32	0.00	-0.00	-4.32	-0.00	0.01
11	12-	13	4.32	-0.00	-0.01	-4.32	0.00	0.00
12	13-	14	5.76	0.00	-0.00	-5.76	-0.00	0.01
13	14-	15	5.76	-0.00	-0.01	-5.76	0.00	0.00
14	15-	16	4.32	0.00	-0.00	-4.32	-0.00	0.01
15	16-	17	4.32	-0.00	-0.01	-4.32	0.00	0.00
16	17-	18	-0.00	-0.00	-0.00	0.00	0.00	0.00
17	1-	10	0.34	0.00	0.00	-0.34	0.00	0.00
18	2-	11	0.00	0.00	0.00	-0.00	0.00	0.00
19	3-	12	0.67	0.00	0.00	-0.67	0.00	0.00
20	4-	13	0.00	0.00	0.00	-0.00	0.00	0.00
21	5-	14	0.67	0.00	0.00	-0.67	0.00	0.00
22	6-	15	0.00	0.00	0.00	-0.00	0.00	0.00
23	7-	16	0.67	0.00	0.00	-0.67	0.00	0.00
24	8-	17	0.00	0.00	0.00	-0.00	0.00	0.00
25	9-	18	0.34	0.00	0.00	-0.34	0.00	0.00
26	1-	11	3.44	-0.00	0.00	-3.44	0.00	0.00
27	11-	3	-2.46	0.00	0.00	2.46	-0.00	0.00
28	3-	13	1.48	-0.00	0.00	-1.48	0.00	0.00
29	13-	5	-0.49	0.00	0.00	0.49	-0.00	0.00
30	5-	15	-0.49	0.00	0.00	0.49	-0.00	0.00
31	15-	7	1.48	-0.00	0.00	-1.48	0.00	0.00
32	7-	17	-2.46	0.00	0.00	2.46	-0.00	0.00
33	17-	9	3.44	-0.00	0.00	-3.44	0.00	0.00

FLUOR DANIEL

SAMPLE DESIGN 1: OPEN DECK CONVEYOR TRUSS



Conveyor Horizontal Truss
(Deflections Shown)



Conveyor Horizontal Truss
Wind Loads